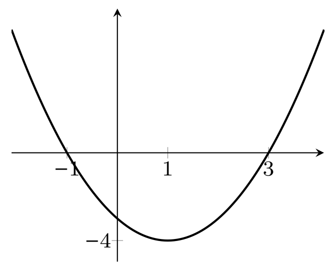
Problem Set 29

The Fundamental Theorem of Calculus, Part 1

1 Let f be the function whose graph is shown below, and let $F(x) = \int_2^x f(t)\,dt$.



(a) Evaluate $F(2)$.

(b) Is $F(3)$ positive, negative, or zero? Explain how you know.

(c) Is $F(1)$ positive, negative, or zero? Explain how you know.

(d) For what values of x is $F(x)$ increasing?

(e) For what values of x is $F(x)$ concave up?

(f) Sketch a rough graph of $F(x)$ on the interval $[-2, 4]$.
Make sure it agrees with everything you've said above!

(g) Now, let $G(x) = \int_3^x f(t)\,dt$. Sketch a rough graph of $G(x)$. (*Hint:* What do you know about how $G(x)$ relates to $F(x)$? What is $G(3)$?)

(h) Suppose that $f(t)$ represents the rate, in students per minute, at which students are entering or leaving Annenberg t minutes after 8 am one morning. If there are 10 students in Annenberg at 8:03

am, sketch the graph of $S(t)$, the number of students in Annenberg t minutes after 8 am.

2 Let $f(x) = \dfrac{2x}{x^2+1}$. Laura is trying to find a formula for the area function $F(x) = \int_1^x f(t)\,dt$.

(a) What is $F(1)$?

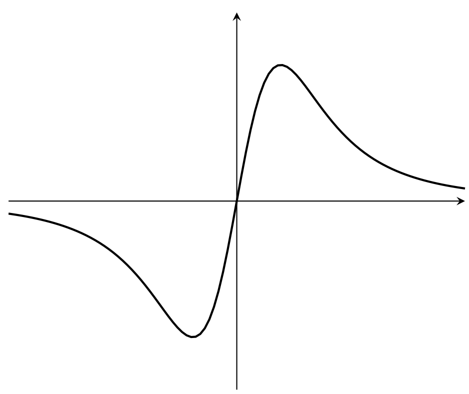
(b) What is $F'(x)$?

(c) One of the following is a formula for $F(x)$; which one? (*Note:* You are **not** expected to be able to directly find a formula for $F(x)$; however, given what you found in the previous parts of this problem, you should be able to eliminate all but one of the choices below.)

A. $\ln(x^2+1)$ B. $\ln(x^2+1) - \ln 2$ C. $\dfrac{2(1-x^2)}{(x^2+1)^2}$ D. $2x \arctan x$

Explain your reasoning.

3 Here is the graph of a function $f(t)$.



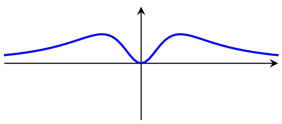
(a) One of the graphs below (choices A–K) is the graph of $F(x) = \int_0^x f(t)\,dt$; which one? Explain your reasoning.

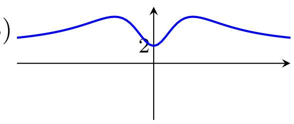
(b) One of the graphs below is the graph of $G(x) = \int_2^x f(t)\,dt$; which one? Explain.

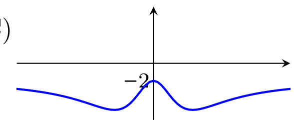
(c) One of the graphs below is the graph of $H(x) = \int_x^2 f(t)\,dt$; which one? Explain.

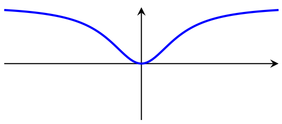
(d) One of the graphs below is the graph of $K(x) = \int_2^3 f(t)\,dt$; which one? Explain.

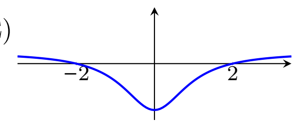
Choices:

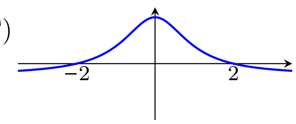
(A) 

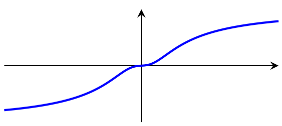
(B) 

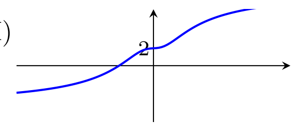
(C) 

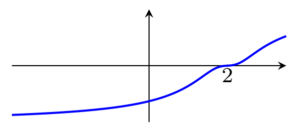
(D) 

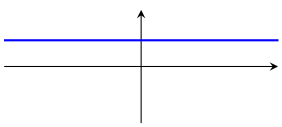
(E) 

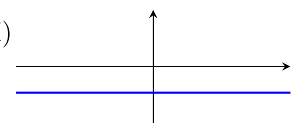
(F) 

(G) 

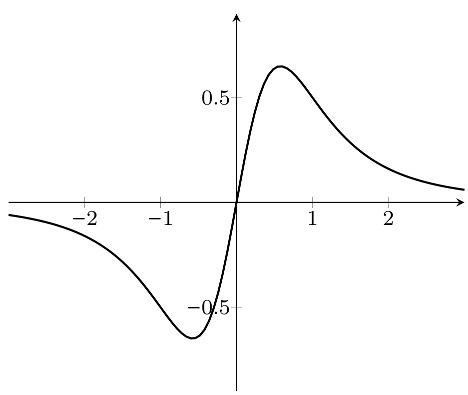
(H) 

(I) 

(J) 

(K) 

4 This problem uses the function $f(t)$ from #3, graphed again below.



Define the functions $F(x) = \int_0^x f(t)\,dt$ and $G(x) = \int_2^x f(t)\,dt$.

(a) Calculate $F'(0)$ and estimate $F'(1)$.

(b) Calculate $G'(0)$ and estimate $G'(1)$.

(c) If we define $J(x) = G(x) - F(x)$, what is $J'(0)$? What is $J'(1)$?

(d) On the graph of $y = f(x)$, shade the region whose area is equal to $|J(0)|$.

(e) Looking at the graphs (A)–(K) in problem #3: which of them could be the graph of $y = J(x)$? Explain your answer fully.